

3) LINEAR METHODS FOR REGRESSION

Assume we have a dataset with 3 predictors (x_1, x_2, x_3) and 1 dependent variable y :

We write the input vector as:

	x_1	x_2	x_3	y
1 sample	1	2	1	3
1 predictor	3	1	5	2
	7	1	2	3

$$X = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 5 \\ 7 & 1 & 2 \end{bmatrix} \text{ Matrix Form}$$

Similarly, the output vector is $Y = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}$

If we have p predictors and n data samples, then, X has dimension $n \times p$ (n rows and p columns). Y has dimension $n \times 1$.

3.2) LINEAR REGRESSION AND LEAST SQUARES

The linear regression model, has form

$$Y = X\beta + E \iff \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1} = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}_{n \times k} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}_{k \times 1} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}_{n \times 1}$$

For a single observation, this becomes:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + E$$

The linear regression model has high bias as it assumes that $E[Y|X]$ is linear (or assuming linearity is reasonable).

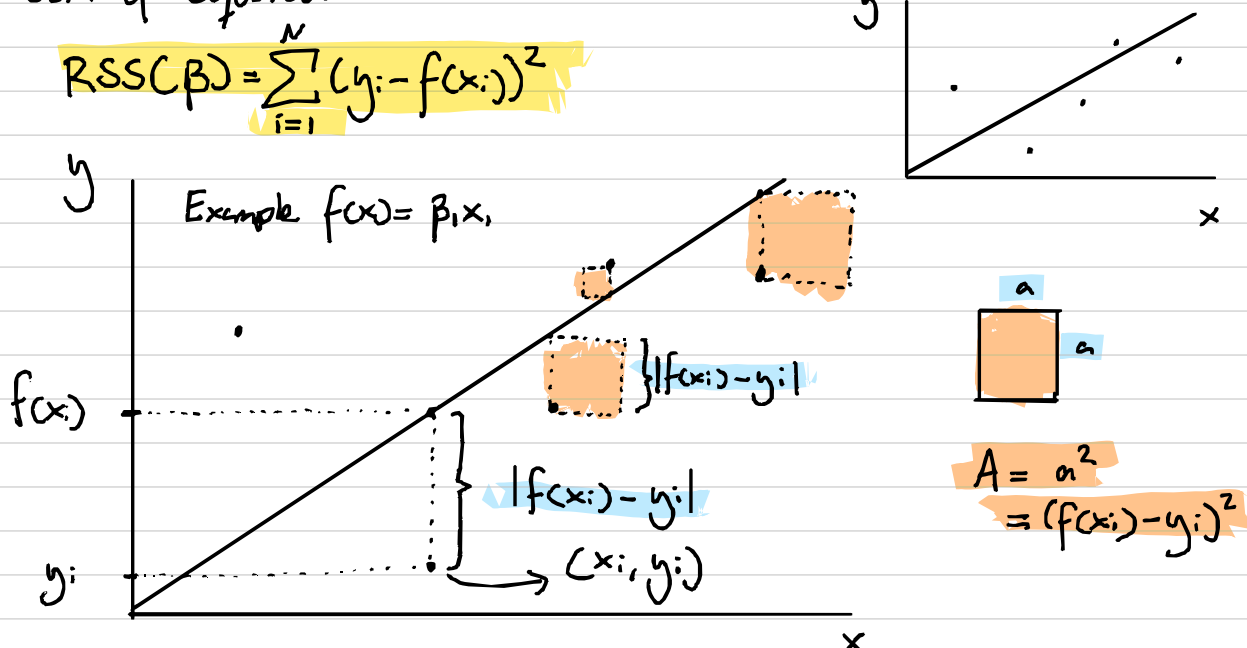
To estimate $\beta = [\beta_0, \beta_1, \dots, \beta_p]^T$ we use a set of training data with N observations $(x_1, y_1), \dots, (x_N, y_N)$:

- $x_i \in \mathbb{R}^p$, $x_i = [x_{i1}, x_{i2}, \dots, x_{ip}]$
- $y_i \in \mathbb{R}$

Let f be our linear model: $f: \mathbb{R}^p \rightarrow \mathbb{R}$
 $x: t \mapsto y:$

Then, the least squares method minimises the residual sum of squares:

$$RSS(\beta) = \sum_{i=1}^N (y_i - f(x_i))^2$$



In matrix form:

$$RSS(\beta) = (Y - X\beta)^T (Y - X\beta)$$

The minimum is obtained by setting:

$$\frac{\partial RSS}{\partial \beta} = 0 \implies -2X^T Y + 2X^T X \beta = 0$$

$$\implies X^T X \beta = X^T Y$$

$$\text{OLS Solution} \implies \hat{\beta} = (X^T X)^{-1} X^T Y$$

*) Assuming that X has full column rank, hence $X^T X$ is invertible and positive definite.

$$\hat{Y} = X \hat{\beta} \quad * \hat{\cdot} \text{ denotes an estimate}$$

$$= X (X^T X)^{-1} X^T Y \quad (\text{From OLS})$$

$$= H Y \quad (\text{Letting } H = X (X^T X)^{-1} X^T)$$

H is sometimes called the hat matrix. Do you see why?

If the columns of X are not linearly independent (e.g. $x_1 = 2x_2$), then β is not uniquely defined. In this case, the OLS formula does not hold. This also happens if $p > N$.

The variance of the estimator is:

$$\text{Var}(\hat{\beta}) = (X^T X)^{-1} \sigma^2, \quad \hat{\sigma}^2 = \frac{1}{N-p-1} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$\hat{\sigma}^2$ is an unbiased estimate of σ^2 . If we assume the conditional expectation of Y ($E[Y|X]$) is linear on $X_1, \dots, X_p$, and Gaussian errors, then.

$$Y = E(Y|X_1, \dots, X_p) + E$$

$$= \beta_0 + \sum_{j=1}^p X_j \beta_j + E \quad E \sim N(0, \sigma^2)$$

Then, $\hat{\beta} \sim N(\beta, (X^T X)^{-1} \sigma^2) \implies$ This can be used to construct hypothesis tests and confidence intervals for β_j

The test hypothesis for a particular coefficient $\beta_j = 0$.

$$\text{Z-score: } z_j = \frac{\hat{\beta}_j}{\hat{\sigma} \sqrt{v_j}} \quad v_j: j\text{-th diagonal element of } (X^T X)^{-1}$$

z_j is distributed as t_{N-p-1} . If σ is known (not estimated), then z_j is normally distributed. For $N-p-1 > 100$ $t_{N-p-1} \sim N$.

F-statistic

$$F = \frac{\frac{RSS_0 - RSS_1}{p_1 - p_0}}{\frac{RSS_1}{n - p_1}}$$

RSS_1 : bigger model
 RSS_0 : smaller model
 p_1, p_0 : n° of predictor (β_0 counts)
 N : n° of training examples

Example

Baseline (model 0): $F(x) = \beta_0 + \beta_1 x_1 + \dots + \beta_4 x_4$

- $RSS_0 = 32.81$ (Residual sum of squares on data)

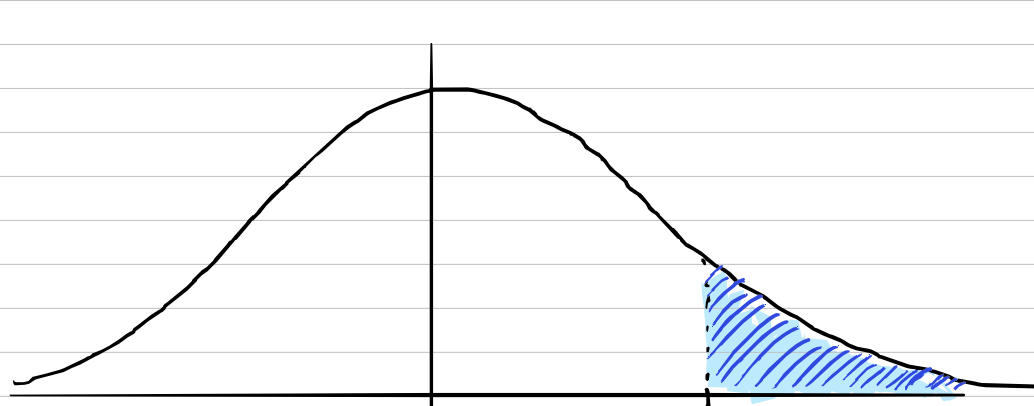
New (model 1): $F(x) = \beta_0 + \beta_1 x_1 + \dots + \beta_8 x_8$ (add 4 predictors)

- $RSS_1 = 29.43$

If both models are trained on 67 data examples, then:

$$F = \frac{(32.81 - 29.43) / (9 - 5)}{29.43 / (67 - 9)} = 1.67$$

$$P(F_{4,58} > 1.67) = 0.17 \implies \text{Not significant}$$



Exercises (Certificate of engagement)

1) Create a linear regression class in Python which:

- Creates a train/test split
- Computes linear regression via OLS
- Computes the residual sum of squares of the model
- Computes the hat matrix
- Computes the variance of the estimate $\hat{\beta}$
- Computes the z-scores of $\hat{\beta}$

2) Create a class to compute the F-statistic between two linear regression models.

3) Use it in the Kaggle playground competition.

3.3) SUBSET SELECTION

- Prediction accuracy
- Interpretation

Objective: Select a subset of $\hat{\beta} = (\beta_0, \beta_1, \dots, \beta_p)$

3.3.1) BEST-SUBSET SELECTION

- Find the subset of size k ($k \in \{0, 1, \dots, p\}$) that gives the smallest residual sum of squares:

$$RSS = \sum_{i=1}^n (y_i - X\beta)^2 \quad \begin{array}{l} \dim(X) = n \times k \\ \dim(\beta) = k \end{array}$$

- Computationally expensive. Why?

How many ways are there of selecting k elements from p elements?

$$\binom{p}{k} = \frac{p!}{k!(p-k)!} \quad n! = n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1$$

→ 'p choose k'

3.3.2) FORWARD- AND BACKWARD-STEPWISE SELECTION

- Sequentially add into the model the predictor that most improves the fit. (Forward)
- Sequentially subtract from the model the predictor that most preserves the fit. (Backward). Works by dropping the lowest z-score.

* Computationally more efficient, greedy algor: lhm, usually similar performance to best subset.

3.4) SHRINKAGE METHODS

Continuous version of subset selection.

3.4.1) RIDGE REGRESSION

- Shrinks coefficients by imposing a penalty on their size.

$$\hat{\beta}_R = \underset{\beta}{\operatorname{argmin}} \left(\sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right)$$

$$\beta = (\beta_1, \beta_2, \dots, \beta_p) - \text{Exclude } \beta_0$$

λ : Shrinkage parameter

Equivalently,

$$\hat{\beta}_R = \underset{\beta}{\operatorname{argmin}} \left(\sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j)^2 \right), \quad \sum_{j=1}^p \beta_j^2 \leq t$$

*) There is a 1-1 correspondence between t & λ .

We normalise the inputs. Can you tell why? How do we exclude β_0 ? Two ways:

1) Remove the column of 1s from X and do:

$$x_j = \frac{X_{ij} - \mu_{x_j}}{\sigma_{x_j}}, \quad y = y - \mu_y$$

$$\boxed{\begin{array}{l} \mu_{x_j} = \frac{1}{N} \sum_{i=1}^N x_{ij}, \quad \sigma_{x_j}^2 = \frac{1}{N} \sum_{i=1}^N (x_{ij} - \mu_{x_j})^2 \\ \mu_y = \frac{1}{N} \sum_{i=1}^N y_i \end{array}}$$

$$\text{Then } \hat{\beta} = (X^T X + \lambda I)^{-1} X^T y \quad (I \text{ has dim } p \times p)$$

$$\hat{\beta}_0 = \mu_y - \mu_x^T \hat{\beta} \quad (\mu_x \text{ has dim } p)$$

2) Change I for $D = \begin{bmatrix} 0 & 0 & \dots \\ 0 & 1 & 0 \\ \vdots & 0 & 1 \end{bmatrix}$, then

$$\hat{\beta} = (X^T X + \lambda D)^{-1} X^T y \rightarrow \beta_0 \text{ included}$$

3.4.2) THE LASSO

$$\hat{\beta}_L = \underset{\beta}{\operatorname{argmin}} \left[\sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j| \right]$$

- No analytical solution, gradient descent?

Exercises (Certificate of engagement)

1) Improve the LinearRegression class from Session 1 to include:

- Best subset selection
- Forward & BACKWARD-Stepwise selection
- Ridge regression

2) Try your class methods on the three datasets:

- What results do you obtain
- Which one works best? Why?

3) Implement the solution for the Lasso using gradient descent (Hard).